Part E — Justification for the e-commerce review figure

Figure: `partD_figure.png`. Script: `partD_figure.py`. Source data: `sessions_hourly.csv` (12,670 rows, Apr 2024 to 11 Sep 2025) and `revenue_catalogue.csv`. All data is synthetic (CDB601220).

1. Each panel, its question, its chart and its channel rule

- **Panel 1, daily sessions.** Question: is daily traffic rising, and how big are the spikes? Line chart, raw in grey with a centred 7-day mean. Channel rule (Week 2): position on a common scale is the most accurate channel for trend, so a line on a zero baseline.
- **Panel 2, weekday by hour heatmap.** Question: when in the week and day are sessions highest? Heatmap with a sequential colormap, because mean sessions has no midpoint (Week 3). Colour carries magnitude only, so the peak is named in the title.
- **Panel 3, treemap of net revenue.** Question: where does the revenue mass sit? Area encodes the value, which is read only to about ±20% (slide 45), so each block prints its value. One hue per category keeps the parent visible.
- **Panel 4, sorted bars.** Question: by how much does each sub-category differ? Position on a shared baseline (Week 2: position beats length beats area), with values printed. It answers the same question as panel 3 in the channel that reads accurately.

2. Time and aggregation decisions

- **Parse.** Two formats, parsed separately with explicit formats. Slash dates are day-first: 2,151 slash rows have a first field above 12, which cannot be a month. The 1,500 ambiguous rows are resolved by format, not guessed. `format="mixed"` was not used.
- **Aggregation.** `sessions` is a flow, so a day is the sum of its filed hours. A day needs at least 21 of 24 hours; 215 days with 21–23 hours are scaled by 24 over the filed hours (a 23-hour day moves by +4.3%). The 13 wholly unfiled days stay NaN and break the line. The day total is never averaged over missing hours.
- **Window: 7 days, centred.** Chosen on the weekly cycle. The 7-day mean removes the day-of-week pattern exactly; the weekly seasonal component has a peak-to-trough of 7,569 sessions per day (partB Q3). It costs 3 days at each end. Centred, because this is a report; the morning dashboard must use a trailing mean, which runs about 15 days late (partB Q4).
- **Partial month.** September 2025 is partial (last filed hour 11 Sep). It is excluded from the analysis, not drawn at full weight.

3. The window declaration

The window (7 days, centred) was fixed before plotting, on the weekly cycle. The series is shown from its first recorded day, 1 April 2024; no start date was selected. Part B Q5 reports every start and length (12 × 5): 33 combinations rise and 27 fall, because short windows on this series depend on where they start. The 28-day version gives the same direction over the full series (slope +21.0 sessions per day against +21.0 for the 7-day). I did not choose the window to strengthen a conclusion; the 7-day choice was made from the cycle.

4. Removed and changed rows

Rule	Count	Effect
Sessions = −1 (not-available code) set to NaN; orders and revenue kept	290	sessions NaN on those hours, not zero
Exact duplicate hours, one copy kept	286	12,670 → 12,384 hours
Hours nobody filed, left absent	312	13 whole days, not filled
Partial September 2025, excluded from analysis	255 filed hours	the series ends 31 Aug 2025
Duplicate SKU-0127 under Wearables, dropped (Audio copy kept)	1	Wearables 174.79 → 159.96 PKR m; share 10.13% → 9.35%
Negative net-revenue rows: kept, not drawn as area	7	sum −1.07 PKR m, netted in sub-category totals

Checks: sub-category shares match `shares_eight.csv` within 0.24 points (tolerance 0.3).

5. Levels rolled up

- **Categories:** 4 (Mobile, Computing, Audio & Imaging, Lifestyle), each drawn as a hue.
- **Sub-categories:** 8, each drawn as a block and a bar. The treemap depth stops here because the eight shares are the object of the comparison, and it sits within the 4 ± 1 chunking limit.
- **Other:** none. Every sub-category is above the 1% threshold (smallest, Drones, 2.4%), so no bucket is hidden. The script prints this as “no categories below 1.0%”.
- **SKU level:** 344 SKUs, not drawn; a treemap of 344 blocks is unreadable.